

Weight of Evidence: From Bletchley Park to Credit Scoring

DENIS BURAKOV

In 1941, Alan Turing turned wartime guesswork into arithmetic. He took odds, logged them, and added tiny scores (decibans) until a hypothesis outweighed its rivals. Eight decades later, banks do the same thing every time you tap “Apply.” Each data point adds or subtracts a small, logarithmically-scaled score; the total crosses a threshold, and a decision falls out. This chapter follows that thread from Turing’s factor method to information theory’s view of evidence, and finally to credit scoring, showing how the same mathematics powers machines that talk, systems that learn, and the scorecards that decide our loans.

1.1 The Birth of Weight of Evidence

The Turing Test and the Challenge of Evidence

Alan Turing: There’s a judge, and a subject. The judge asks questions, and based on the subject’s answers, he determines: Who is he speaking with? What is he speaking with? All you have to do is ask me a question.

— Graham Moore, *The Imitation Game* (2014)

The Turing Test poses a fundamental question about judgment: How do we accumulate evidence to make a decision? How many questions must the judge ask? How much certainty is enough? These aren’t merely philosophical puzzles; they’re practical problems that Turing confronted during World War II, when the stakes were measured in lives rather than academic debates.

Between 1941 and 1943, at Bletchley Park, Turing faced his own version of this challenge. Instead of judging whether a hidden subject was human or machine, he needed to judge which of millions of possible Enigma settings had encrypted the day’s German naval messages. He could not test them all as there was not enough time. He needed a method to accumulate evidence into a score, and to know when he had gathered “enough” information to make a decision.

His solution was Banburismus, a deceptively simple scoring system based on Bayesian inference principles. The techniques Turing developed to break Nazi codes would remain classified for seventy years. When they finally emerged, the world discovered something remarkable: the same mathematical principles that helped win the war had been independently rediscovered and were already powering spam filters, medical diagnostics, and credit scoring systems.

Banburismus: Turing’s Bayesian Solution

During the active phase of World War II, between 1941 and 1943, Turing developed a manual method for breaking Enigma codes that relied on pattern recognition and probability [13].

On a given day, Turing’s team would sit at long tables in Hut 8, aligning Banbury sheets, scoring matches, and accumulating decibans until one rotor setting outweighed the rest. His team, composed primarily of women assistants, used specially printed paper with holes punched for letters, which could be manually aligned to look for repeated patterns between pairs of messages. These patterns, in turn, suggested common settings of the rotors in the Enigma machine [13].

The paper was printed in Banbury, and the entire process became known as *Banburismus*. The aim of “banburizing” was to suggest plausible inputs to the electromechanical Bombe computer, and to rule out other settings in order to reduce the time spent on Bombe runs [13].

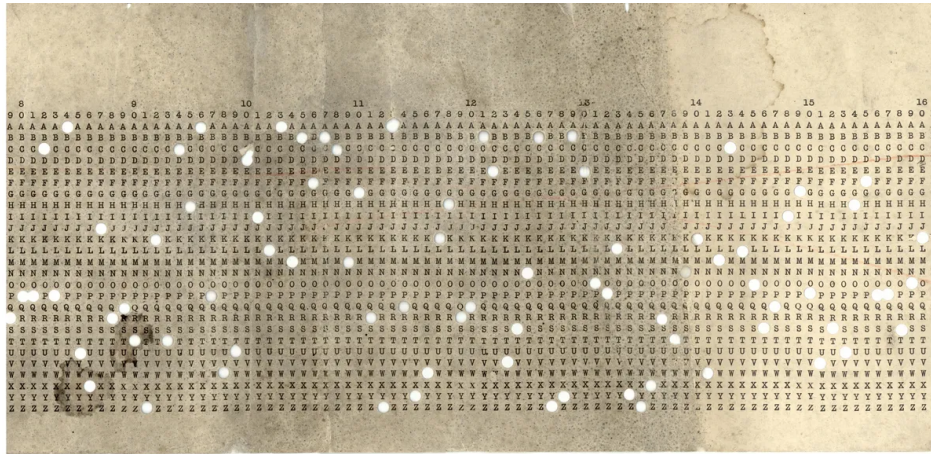


Figure 1.1: An original Banbury sheet used for early attacks on the naval Enigma, found during the restoration of Bletchley Park huts. Courtesy of Bletchley Park Trust.

Turing wanted to use Bayes’ theorem to examine the relative probability for competing hypotheses about the underlying settings of the Enigma machine used to generate the coded message. He made two innovations to make the computation feasible: first, rather than repeatedly using Bayes’ theorem in its odds form, which requires multiplying likelihood ratios, he took logarithms to the base 10 and so reduced the problem to adding $\log(\text{likelihood ratios})$. Second, he multiplied the $\log(\text{likelihood ratios})$ by 10 (later 20), and then rounded the result to a whole number. So the whole process was reduced to adding and subtracting whole numbers, which

could be carried out with pencil and paper [13].¹

From Multiplication to Addition: The Logarithmic Transform

The mechanical process of Banburismus was elegant, but its mathematical foundation was revolutionary. Turing's breakthrough was to transform Bayesian inference from a multiplicative process into an additive one by taking logarithms of likelihood ratios and scaling them to whole numbers for manual computation. The units for these scaled logarithms became known as “decibans” (dB) and “half-decibans” (hdB). According to Jack Good, the half-deciban represents approximately the smallest change in weight of evidence that is directly perceptible to human intuition [13].

As Turing described in 1941 [13]:

“The probability of an event on certain evidence is the proportion of cases in which that event may be expected to happen given that evidence.”

Log-likelihood ratios had previously been called ‘weights of evidence’ by philosopher C.S. Peirce in 1878 [6]. Turing was apparently unaware of this precedent, but Jack Good later reintroduced the term, establishing the connection between Turing's wartime work and the broader philosophical tradition [6].

Turing wrote a paper describing this work, but it remained classified for seventy years. When it finally appeared in 2012, a modern government codebreaker (known only as ‘Richard’) said the material could be released as they had ‘squeezed the juice’ out of it [13]. Meanwhile, Turing's technique had been independently rediscovered and become a standard part of machine learning—the “naive Bayes” classifier used in spam detectors and medical diagnostic systems [10].

The Power of Additive Scoring

The transformation from multiplicative to additive operations fundamentally changed how evidence could be accumulated. As Hugh Alexander noted [1]:

“We found that by splitting up a job into as many separate parts as possible and by having a really adequate set of scoring tables...the whole process [was] reduced to a matter of looking things up in a number of different tables and the only mathematical operations involved were addition and subtraction.”

This approach, which one historian described as evidence that “there was no time for calculating individual probabilities. Instead, Hut 8 devised a streamlined process of scores that required only to be added or subtracted by non-mathematical (but clever) ‘girls’” [1], represented a revolutionary approach to inference under constraints.

The arithmetic was needed for deciding how much evidence was “enough,” that is, sufficient to decide that a particular speculative alignment could be relied upon before using it in the

¹Throughout this chapter, when we write log without specifying a base, we mean the natural logarithm (base e), following standard mathematical convention in statistics and machine learning literature. Turing's original work used base 10 logarithms for computational convenience.

complex reconstruction process that was to follow [1]. Neither of the contemporaneous histories explains the derivation of this arithmetic explicitly, though it was based on Bayes' theorem and the calculation of mathematical probabilities of different numbers of repeats [1].

The process was called “decibanning,” and as one observer noted, “Banburismus is about the best example of the use of scoring systems the writer has seen. It would have been quite impossible without this technique, and the more whole-heartedly it was employed, the better the results obtained” [14].

Named for the Oxfordshire town where the sheets were printed, measures of probability were called ‘bans’ or ‘decibans’, using a logarithmic scale to accumulate evidence. The process of adjusting hypotheses based on small observations was, as one historian wryly noted, “distressingly Bayesian” [14].

The Cambridge mathematician I.J. Good joined Bletchley Park in April 1940 and was assigned to Hut 8 [14]. He is reported to have asked Turing, ‘Aren’t you essentially using Bayes’s theorem?’ and Alan confessed that it was so [14]. After the war, Good became a professional defender of Bayesian methodology, writing several papers on the subject; he had to do so from the safety of the United States [14].

The Logic of Judgment

The principle of accumulating evidence through additive scoring captures something fundamental about human reasoning. Long before Turing formalized the mathematics, people intuitively understood that judgment involves weighing multiple pieces of evidence until confidence crosses some threshold.

Consider Bulgakov’s portrayal of Pontius Pilate interrogating Yeshua Ha-Nozri in *The Master and Margarita*. Pilate’s decision-making process mirrors Bayesian inference: each question and answer shifts his assessment incrementally. Yeshua’s calm demeanor provides evidence for innocence; the priests’ accusations argue for guilt; the crowd’s hostility adds noise rather than signal. Pilate accumulates this evidence mentally, his confidence shifting with each exchange, until political necessity forces a decision.

This intuitive process—accumulating evidence until a threshold is crossed—appears across human judgment contexts. British courts recognize it implicitly: individual pieces of evidence are admissible, but juries cannot systematically multiply likelihood ratios. The law insists that combining evidence must remain human judgment, not mathematical formula.

Turing’s breakthrough was making this implicit process explicit. At Bletchley Park, he converted judgment into arithmetic: evidence could be accumulated systematically, scores could be tallied, and decisions could be reached when thresholds were crossed. What had been intuitive became mechanical, what had been subjective became reproducible.

1.2 The Mathematical Foundation

Having seen how Banburismus worked in practice, we can now examine the mathematical principles that made it so effective.

The Factor Principle: Bayes' Theorem in Logarithmic Form

The mathematical elegance of Banburismus lies in its systematic treatment of hypothesis testing through what Turing called the Factor Principle [1]. This principle provides a framework for accumulating evidence that remains fundamental to modern statistical inference and machine learning.

We start with a hypothesis to be tested, for example, that a certain statement is true. We also start with some understanding of the credibility of the hypothesis, which may be objective or subjective. We can express that credibility in terms of odds. Odds are straightforward: if something happens on average four times out of five trials (corresponding to an 80% probability), the odds are 4:1 in favor of its happening (or 1:4 against its not happening) [1].

What we start with are the *prior odds*, prior, that is, to an event that gives us some additional evidence. We now need to know whether that event is more likely to happen when the hypothesis is true or when it is false, and how much more likely. More precisely, we need the ratio of these two probabilities. This ratio is called the “Bayes factor” or, in Turing’s terminology, “the factor” [1].

Bayes’ theorem tells us that the credibility after we have the additional evidence (the *posterior odds*) is obtained by multiplying the prior odds by the Bayes factor:

$$\text{Posterior odds} = \text{Prior odds} \times \text{Bayes factor} \quad (1.1)$$

where

$$\text{Bayes factor} = \frac{P(\text{evidence} | \text{hypothesis true})}{P(\text{evidence} | \text{hypothesis false})} \quad (1.2)$$

The credibility is enhanced or diminished according to whether the factor is above or below 1. Moreover, if there are successive pieces of evidence that are independent of each other, their factors can be multiplied to give a composite factor [1]. This multiplicative property is the foundation of what we now call the naive Bayes classifier [10].

This relationship can also be written in the formal “odds form” of Bayes’ theorem:

$$O(H | E) = O(H) \times \frac{P(E | H)}{P(E | \bar{H})},$$

where $O(H | E)$ and $O(H)$ denote the posterior and prior odds respectively, and $\frac{P(E|H)}{P(E|\bar{H})}$ is the *Bayes factor* or, in Turing’s terms, the *factor in favor of H provided by E* [3].

Turing’s innovation was to work with logarithms of these factors. Taking logarithms of both sides:

$$\log(\text{Posterior odds}) = \log(\text{Prior odds}) + \log(\text{Bayes factor}) \quad (1.3)$$

This transformation converts multiplication into addition, making the calculations tractable for manual computation. The $\log(\text{Bayes factor})$ became known as the “weight of evidence.”

Information Theory and Evidence

In 1948, Claude Shannon formalized the concept of information in probabilistic terms in his landmark paper “A Mathematical Theory of Communication” [7], defining it as the negative

logarithm of probability, the same mathematical operation that underlies Turing’s weight of evidence. Shannon’s information content measures how surprising an event is; Turing’s weight of evidence measures how strongly an observation shifts belief. Both quantify change in knowledge using logarithms of probability. Where Turing’s concern was inference (deciding between hypotheses), Shannon’s was communication, i.e. distinguishing messages. Their mathematics, however, is identical. Later writers such as I.J. Good and David MacKay would make this unity explicit: inference is information processing.

The connection between Banburismus and modern information theory is profound [5]. When we compare hypotheses with each other in the light of data, it is often convenient to compare the log of the probability of the data under the alternative hypotheses:

$$\log \text{evidence for } H_i = \log P(D|H_i) \quad (1.4)$$

Or, in the case where just two hypotheses are being compared, we evaluate the ‘log odds ratio’:

$$\log \frac{P(D|H_1)}{P(D|H_2)} \quad (1.5)$$

This quantity is the ‘weight of evidence in favor of H_1 ’ [5]. The log evidence for a hypothesis, $\log P(D|H_i)$, is the negative of the information content of the data D under that hypothesis: if the data have large information content given a hypothesis, then they are surprising to that hypothesis; if some other hypothesis is not as surprised by the data, then that hypothesis becomes more probable [5].

As David MacKay elegantly notes, ‘information content’, ‘surprise value’, and log-likelihood (or log evidence) are the same thing [5]. This unification connects Shannon’s information theory [7] with Bayesian inference and provides the theoretical foundation for modern machine learning algorithms.

Formally, information theory quantifies the notion of “surprise.”

The *information content* (or self-information) of an outcome x , with probability $P(x)$, is defined as

$$h(x) = -\log P(x). \quad (1.6)$$

Unlikely events (small $P(x)$) have high information content, meaning they are more surprising.

To understand this intuitively, consider a coin that lands heads 90% of the time [11]. We can express the surprise of an outcome as $1/P(x)$, but Shannon showed it is better to define it as the logarithm: $\log(1/P(x)) = -\log P(x)$. This logarithmic measure, known as Shannon information or *surprisal*, has the crucial property that surprises from independent events can be added rather than multiplied.

The *entropy* of a random variable X , measuring the expected surprise across all outcomes, is defined as

$$H(X) = -\sum_x P(x) \log P(x). \quad (1.7)$$

Entropy represents the average information content in a random variable X . In essence, entropy measures uncertainty: when our uncertainty is reduced, we gain information. Information and entropy are thus two sides of the same coin.

The choice of logarithm base determines the units of measurement. Using natural logarithms (base e) gives units called *nats*. Using base-10 logarithms gives units called *bans*, named after the English town of Banbury where the special cards used at Bletchley Park were printed. This historical connection reinforces the deep relationship between Turing's wartime work and Shannon's information theory.

In the context of weight of evidence, log-likelihood ratios compare the relative surprises of data under competing hypotheses. A hypothesis that finds the data less surprising accumulates more weight of evidence in its favor.

The Historical Context and Legacy

Neither Alexander’s nor Mahon’s contemporaneous histories mentioned the word ‘Bayes’, and Turing himself generally referred to “the factor” [1]. I.J. Good also recalls that: “When I said to him that the concept was essentially an application of Bayes’s theorem he said ‘I suppose so’ ” [4]. In Turing’s later declassified paper [13], we can see Factor Principle (Bayes’ Theorem).

The historical significance of the ban as a unit of measurement cannot be overstated [5]. When Alan Turing and the other codebreakers at Bletchley Park were breaking each new day's Enigma code, their task was a massive inference problem: to infer, given the day's cyphertext, which three wheels were in the Enigma machines that day; what their starting positions were; what further letter substitutions were in use on the steckerboard; and, not least, what the original German messages were [5].

These inferences were conducted using Bayesian methods, and the chosen units were decibans or half-decibans, with the deciban judged to be the smallest weight of evidence discernible to human intuition [5]. The evidence in favor of particular hypotheses was tallied using sheets of paper that were specially printed in Banbury. The inference task was known as Banburismus, and the units in which Banburismus was played were called bans, after that town [5].

Using increasingly sophisticated language models, the evidence contributed by runs of matches was more accurately computed [5]. Such a scoring system was worked out by Turing and refined by Good. Positive results were passed on to automated and human-powered codebreakers. According to Good, the longest false-positive that arose in this work was a string of 8 consecutive matches between two messages that were actually in unrelated states [5], a testament to the robustness of the methodology.

MacKay's Reconstruction of Banburismus

To illustrate how Turing’s method works in practice, David MacKay reconstructed a simplified version of Banburismus using two English nursery rhymes [5]. Consider comparing these text strings:

```
u:      LITTLE-JACK-HORNER-SAT-IN-THE-CORNER-EATING-A-CHRISTMAS-PIE--HE-PUT-IN-H
v:      RIDE-A-COCK-HORSE-TO-BANBURY-CROSS-TO-SEE-A-FINE-LADY-UPON-A-WHITE-HORSE
matches: .*...*.******.*.....*.....*.....*.....*
```

The question: Are these two strings related (hypothesis H_1) or unrelated (hypothesis H_0)? The bottom row shows matches (*) and non-matches (.) between u and v .

Breaking this sequence into segments:

Position 0-19: u : LITTLE-JACK-HORNER-S
 v : RIDE-A-COCK-HORSE-TO
 $\downarrow$: .*...*...*****.*...

Position 20-39: u : AT-IN-THE-CORNER-EAT
 v : -BANBURY-CROSS-TO-SE
 $\downarrow$:*.....

Position 40-59: u : ING-A-CHRISTMAS-PIE-
 v : E-A-FINE-LADY-UPON-A
 $\downarrow$: ...*.....

Position 60-71: u : -HE-PUT-IN-H
 v : -WHITE-HORSE
 $\downarrow$: *......

MacKay's example compares each character position, accumulating evidence:

- **Character match:** +3.01 decibans (log-evidence for H_1 per match)
- **Character mismatch:** -0.18 decibans (log-evidence for H_1 per non-match)

Key Matches Identified: The algorithm finds 12 character matches, including:

- Position 6: '-' (dash) in both strings
- Positions 9-14: 'CKHOR' sequence in "JACK-HORNER" vs "COCK-HORSE"
- Position 31: 'O' in "CORNER" vs "CROSS"
- Position 43: '-' (dash) in both strings
- Position 60: '-' (dash) in both strings

Starting from neutral (0 decibans), the algorithm finds 12 matches and 60 non-matches:

$$\begin{aligned} \text{Total evidence} &= 12 \times (+3.01) + 60 \times (-0.18) & (1.8) \\ &= 36.12 - 10.8 = 25.32 \text{ decibans} & (1.9) \end{aligned}$$

Converting to odds: $10^{25.32/10} \approx 354 : 1$ in favor of the strings being related.

This demonstrates the power of Turing's additive approach: even though most character positions don't match (60 vs. 12), the few matches provide such strong evidence that they overwhelm the many weak counter-indications. The algorithm correctly identifies that both strings are English nursery rhymes, despite their surface differences.

The same mathematical principle, accumulating logarithmic evidence until a threshold is crossed, powers modern credit scoring systems.

1.3 Weight of Evidence in Credit Scoring

The mathematical principles developed at Bletchley Park didn't remain locked in wartime archives. They found new life in an unexpected domain: consumer finance.

From Codebreaking to Credit Decisions

Today, data scientists working on credit scoring face remarkably similar challenges to those faced by Turing and his team. Both domains require:

1. **Accumulating evidence** from multiple sources (like Turing's repeated patterns in encrypted messages)
2. **Handling computational constraints** (though modern constraints differ from wartime pencil-and-paper calculations)
3. **Making decisions under uncertainty** (like determining which Enigma settings to test)
4. **Balancing speed and accuracy** (like the need to break codes before they became obsolete)
5. **Working with features that may or may not be independent** (requiring careful modeling choices)

The scoring systems developed by Turing and refined by Good represent some of the most sophisticated early examples of what we now recognize as machine learning. The mathematical principles remain unchanged, even as the computational tools have evolved dramatically.

From Decibans to WOE: The Mathematical Bridge

To understand how Turing's wartime innovations connect to modern credit scoring, we need to examine the Weight of Evidence (WOE) transformation, a staple of credit risk modeling that has its roots directly in the Factor Principle.

Weight of Evidence quantifies the strength of evidence supporting or refuting a hypothesis, derived from independent pieces of data. The mathematical foundation, drawn directly from Turing's declassified 1941 paper [13], is:

$$\begin{aligned} \text{A posteriori odds of the theory} &= \text{A priori odds of the theory} \\ &\times \frac{\text{Probability of the data being fulfilled if the theory is true}}{\text{Probability of the data being fulfilled if the theory is false}} \end{aligned} \quad (1.10)$$

In this formula, the *theory* is that the man died of heart failure, and the *data* is that he died in his bed. The general formula above is described as the *factor principle*. The ratio

$$\frac{\text{Probability of the data if the theory is true}}{\text{Probability of the data if the theory is false}} \quad (1.11)$$

is what Turing called the factor for the theory on account of the data. This factor later became known as the Bayes factor [1].

To illustrate this principle, consider the example that Turing himself provided in his classified 1941 paper [13]. The theory is that a man died of heart failure, and we have three pieces of evidence:

Evidence in favor of heart failure:

- $2/3$ of men who die of heart failure die in their beds
- $2/5$ of those who die of heart failure have fathers who died of heart failure
- $1/2$ of those who die of heart failure have bedrooms on the ground floor

Evidence from other causes:

- $1/4$ of men who die from other causes die in their beds
- $1/6$ of those who die from other causes have fathers who died of heart failure
- $1/20$ of those who die from other causes have bedrooms on the ground floor

The final factor in favor of the heart failure hypothesis, Turing demonstrates, is the product of three factors: $(2/3)/(1/4)$, $(2/5)/(1/6)$, and $(1/2)/(1/20)$, arising from the three independent pieces of evidence. The interpretation of each factor is that the credibility in the hypothesis is enhanced or diminished according to whether the factor is above or below 1. In total, we obtain odds of 64 to 1 in favor of the heart failure hypothesis (without incorporating prior odds).

Turing's calculation for the first factor shows that dying in bed is 2.67 times more likely given heart failure than given other causes, yielding a Bayes factor of $8/3$. The derivations of the remaining two factors in Turing's example [13] follow the same reasoning and are left to the reader.

Alan Turing described the *weight of evidence* as the logarithm of the factor by which the odds of a theory change when new data arrive. He used *decibans* ($\log_{10}$ scale $\times 10$) to quantify the strength of evidence.

Step	Description	Expression	Value
1	A priori odds of the theory (heart failure)	$(1/5) / (4/5)$	0.25
2	Factor for the theory on account of the data (in bed)	$(2/3) / (1/4)$	2.66
3	A posteriori odds of the theory	0.25×2.66	0.67

Table 1.1: Turing's Factor Principle Applied to Heart Failure Diagnosis

In words:

- Before seeing the data, the theory of heart failure has odds of 1:4 (0.25).
- Being found *in bed* increases the odds in favor of heart failure by a factor of 2.66.
- The updated (posterior) odds are 0.67, or roughly 2:3.

Turing's scoring (in decibans):

$$\text{Score} = 10 \log_{10}(2.66) = 4.3 \quad (1.12)$$

“We score 4.3 decibans in favor of the heart-failure theory.”

The logarithmic factor is the **Weight of Evidence (WOE)**.

For scoring purposes, Turing proposed using the base-10 logarithm of factors to convert them to “decibans,” named after the English town of Banbury [13]. Decibans were adopted for faster manual computation because scores could be added rather than multiplied, and because whole numbers are easier to work with than fractional probabilities.

This is precisely the same transformation that modern credit scoring systems use. In credit risk modeling, we call $\log(\text{Bayes factor})$ the “Weight of Evidence” (WOE). Turing’s work lays the foundation for modern scoring systems, where each input contributes an independent piece of evidence to the final decision. Logarithmic scoring models assign WOE values to features, with the final score representing the sum of log factors that reflect cumulative evidence. This method forms the backbone of credit scoring as we know it today.

Weight of Evidence in Practice: A Credit Scoring Example

To see how this works in practice, consider a group of bank customers whose credit card utilization (the percentage of available credit being used) exceeds 90%. How many of these customers are likely to default on their loans?

A contingency table for this scenario shows the relationship between credit utilization and default status, where high utilization (>90%) is strongly associated with default².

Step	Expression	Value
A priori odds of default	$(1/10) / (9/10)$	0.11
Factor (likelihood ratio)	$(4/5) / (1/10)$	8.0
A posteriori odds	0.11×8.0	0.88
A posteriori probability of default	$0.88 / (1 + 0.88)$	0.47

Table 1.2: Credit Scoring Example: High Utilization and Default Risk

Interpretation:

- Before observing usage, the base odds of default are 1:9 (≈ 0.11).
- High usage (>90%) increases odds by $\times 8$.
- Posterior probability of default $\approx 47\%$.

Weight of Evidence (natural log):

$$\text{WOE} = \log(8.0) = 2.0794 \quad (1.13)$$

Logit form:

²Some credit risk texts use a “good-to-bad” WOE formulation [8], likely to align with traditional scorecards where higher scores indicate better creditworthiness. However, this can create mathematical inconsistency with the model intercept, which represents baseline log-odds of default. The standard “bad-to-good” formulation used here ensures positive WOE values indicate increased risk and negative values indicate decreased risk.

Component	Expression	Value
Log prior odds	$\ln(0.11)$	-2.2073
Add WOE	$-2.2073 + 2.0794$	-0.128
PD (logistic)	$\sigma(-0.128) = \frac{e^{-0.128}}{1+e^{-0.128}}$	0.468

Table 1.3: Logit Transformation of WOE Value

So, the predicted probability of default is **47%**, consistent with the odds-based computation.

If we examine the group with utilization exceeding 90%, we find 80 of the 100 customers who defaulted and 90 of the 900 customers who did not default. The Bayes factor (likelihood ratio) is then:

$$\text{Odds of Default | Usage} > 90\% = \frac{80/100}{90/900} = \frac{0.80}{0.10} = 8.0 \quad (1.14)$$

This means that in this high-utilization group, customers are 8 times more likely to be defaulters than non-defaulters. The Weight of Evidence value is then:

$$\text{WOE} = \log(8.0) = 2.08 \quad (1.15)$$

To convert this to a probability, we must combine the WOE with the prior log odds. The prior probability of default in this dataset is $100/1000 = 0.10$, giving prior odds of 100:900 or 1:9. The prior log odds are:

$$\log(\text{prior odds}) = \log(1/9) = -2.20 \quad (1.16)$$

Adding the WOE value to the prior log odds:

$$\log(\text{posterior odds}) = -2.20 + 2.08 = -0.128 \quad (1.17)$$

Passing this value through the logistic (sigmoid) function yields a probability:

$$P(\text{default} | \text{utilization} > 90\%) = \frac{e^{-0.128}}{1 + e^{-0.128}} = 0.47 \quad (1.18)$$

We can verify this result by calculating the proportion of defaulters directly within the high-utilization group: $80/(80+90) = 0.47$. The Bayesian approach and the frequentist calculation yield identical results, demonstrating the mathematical consistency of the framework.

The Two Faces of Weight of Evidence

It is crucial to understand that WOE alone cannot determine probability; it must always be interpreted relative to the prior log odds. The final probability is obtained by aggregating the sum of WOE values across all features and combining them with the prior log odds. This additive structure is what makes WOE-based scoring systems so computationally efficient and interpretable.

Each WOE value represents a piece of evidence for or against default, independent of other features. At Bletchley Park, the focus was primarily on the values themselves, with decision

thresholds set empirically. However, to derive rigorous probability estimates, we must combine WOE values with prior odds, as demonstrated above. WOE acts as an adjustment added to the prior log odds to yield the posterior log odds and, ultimately, the final probability estimate.

In most modern applications, logistic regression is used as the final step to compute probabilities. Maximum likelihood estimation assigns weights to WOE-transformed features to produce the final logit and derive probability estimates. However, when assuming the independence of features (the naive Bayes assumption) probabilities can be derived directly by summing the WOE values across features without requiring logistic regression. This is precisely analogous to how Turing accumulated decibans to reach decision thresholds.

We can calculate WOE in two equivalent ways. The first approach uses a contingency table (as shown above), while the second leverages the Bayes factor directly (see Jack Good's explanation [4]). Both methods yield identical results—in this case, a factor of 8.0 for high credit utilization.

We can express this reasoning directly from a **2×2 contingency table**:

Credit Usage	Default = 0	Default = 1	Total
≤ 90%	810	20	830
> 90%	90	80	170
Total	900	100	1000

Table 1.4: Contingency Table: Credit Utilization vs Default

Variant 1 — Conditional Probability Table

Credit Usage	P(Default=0)	P(Default=1)
≤ 90%	0.90	0.20
> 90%	0.10	0.80
Total	1.00	1.00

Table 1.5: Conditional Probability Table

Factor in favor of default:

$$\frac{0.8/0.1}{0.2/0.9} = 8.0 \quad (1.19)$$

$$\text{WOE} = \log(8.0) = 2.08 \quad (1.20)$$

Variant 2 — Using Default Rates (The Odds Form of the Bayes' Theorem)

$$P(\text{default} | > 90\%) = 0.471, \quad P(\text{no default} | > 90\%) = 0.529 \quad (1.21)$$

Credit Usage	Default rate
$\leq 90\%$	2.4%
$> 90\%$	47%
Total	10%

Table 1.6: Default Rates by Credit Usage

$$\text{Prior odds} = \frac{0.10}{0.90} = 0.111 \quad (1.22)$$

$$\text{Posterior odds} = \frac{0.471}{0.529} = 0.891 \quad (1.23)$$

$$\text{Factor} = \frac{0.891}{0.111} \approx 8.0 \quad (1.24)$$

$$\text{WOE} = \log(8.0) = 2.07 \quad (1.25)$$

“Credit usage $>90\%$ increases the odds of default by a factor of $8\times$.”

This is the same *weight of evidence* that Good formalized via the odds form of Bayes’ theorem, written as:

$$O(H | E) = O(H) \times \frac{P(E | H)}{P(E | \bar{H})}$$

or equivalently,

$$\frac{O(H | E)}{O(H)} = \frac{P(E | H)}{P(E | \bar{H})},$$

where the ratio on the right-hand side is the *factor in favor of H provided by E* , also known as the *Bayes factor* or *Turing–Good weight of evidence*.

The Independence Assumption and Its Limitations

The mathematical foundation for combining independent evidence has an interesting history. Charles Sanders Peirce introduced the concept in 1878, showing how to add weights of evidence from independent sources. However, Peirce made a subtle but crucial error: he summed evidence without properly accounting for prior odds. This oversight wasn’t corrected until Turing and Good’s wartime work established the rigorous mathematical framework that underlies modern additive scoring systems (see bibliographical notes for the full story).

The naive Bayes assumption (that features contribute independent evidence) is both the great strength and the primary limitation of WOE-based scoring. This assumption allows for simple additive scoring and makes models highly interpretable, but this assumption breaks down when features are correlated or interact in complex ways.

Modern credit scoring has evolved beyond pure WOE models to incorporate methods that can capture feature interactions: gradient boosting machines, neural networks, and ensemble

methods. Yet WOE transformations remain valuable for several reasons: they provide interpretable feature contributions, enable regulatory compliance through explainable scoring, and offer computational efficiency in production systems.

As D. Spiegelhalter [9] has noted, Alan Turing’s approach to explanation during Enigma codebreaking relied fundamentally on weights of evidence as a method of communication: log-likelihood ratios implemented through a naïve Bayes model. Jack Good later promoted this methodology in the 1950s, and the additive scoring approach pioneered in Turing’s wartime work has become fundamental to modern machine learning [10].

The naïve Bayes classifier expresses posterior odds as the product of individual likelihood ratios:

$$\frac{p(H_0 \mid s_1, \dots, s_p)}{p(H_1 \mid s_1, \dots, s_p)} = \prod_i \frac{p(s_i \mid H_0)}{p(s_i \mid H_1)} \times \frac{p(H_0)}{p(H_1)} \quad (1.26)$$

Taking logarithms transforms this into the additive form that Turing pioneered:

$$\log \frac{p(H_0 \mid s_1, \dots, s_p)}{p(H_1 \mid s_1, \dots, s_p)} = \sum_i w_i + \log \frac{p(H_0)}{p(H_1)} \quad (1.27)$$

where w_i is the *weight of evidence*

$$w_i = \log \frac{p(s_i \mid H_0)}{p(s_i \mid H_1)} \quad (1.28)$$

- Weights of evidence are positive/negative if evidence s_i is for/against H_0
- Multiplying by 10 and rounding helps explanation

Even with its independence assumption, the naïve Bayes framework remains one of the most interpretable probabilistic models available. By translating multiplicative likelihoods into additive scores, Turing and Good effectively created the first explainable model, a design still mirrored in every WOE-based scorecard used in consumer lending.

The Enduring Legacy

The journey from Bletchley Park’s codebreaking efforts to modern credit scoring is not merely a historical curiosity; it represents a direct lineage of mathematical and computational thinking. The principles that helped defeat Enigma (accumulating evidence through logarithmic scoring, making decisions under uncertainty, and balancing accuracy with computational efficiency) remain central to how banks make lending decisions today.

The methods that helped win World War II have evolved into the algorithms that help us make better decisions about risk, opportunity, and the future. In the chapters that follow, we’ll explore how these fundamental principles extend to modern machine learning methods, examining both where the simple additive structure of WOE remains sufficient and where more complex models become necessary.

What began as pencil-and-paper calculations in a wartime codebreaking facility has become the mathematical foundation for algorithms that process millions of credit applications annually. The deciban, once the smallest unit of evidence perceptible to human intuition, has evolved into

the weights and activations of neural networks. Yet the underlying principle remains unchanged: evidence accumulates, scores rise and fall, and when the total crosses a threshold, a decision emerges.

Today, the ideas underlying weight of evidence extend far beyond credit scoring. While not always used explicitly, the same logic of combining independent pieces of probabilistic evidence underpins modern fraud detection systems, medical diagnostics, recommendation algorithms, and risk models across industries. From spam filters to autonomous vehicles, the mathematical framework Turing developed to break enemy codes now helps machines make sense of an uncertain world.

Bibliographical and Historical Notes

Ronald A. Fisher and Alan M. Turing

Among the many figures who shaped twentieth-century science, few had a deeper influence on the foundations of modern data analysis than Ronald A. Fisher and Alan M. Turing. Both were British prodigies who transformed how knowledge could be extracted from uncertainty. Fisher formalized the logic of inference, turning statistics into an exact science, while Turing redefined reasoning itself through computation and probability. Though they never collaborated directly, their intellectual legacies form complementary parts of a single story.

Ronald A. Fisher and Alan Turing crossed paths in the mid-1930s when Turing submitted his fellowship dissertation at King's College, Cambridge. Fisher, already a leading figure in statistics, was one of the examiners. He found the subject unappealing and dismissed it as overly explored, but nevertheless acknowledged the originality of Turing's methods and endorsed him as a candidate of real distinction. It was an early test of Turing's academic path, and Fisher's reluctant approval helped secure his place at Cambridge to be elected to a fellowship of King's at the ripe old age of 22 on 16 March 1935 [12].

Their intellectual trajectories later diverged in striking ways. Fisher was a staunch defender of orthodox statistical methods (frequentist) and a vehement critic of Bayesian approaches making Bayesian analysis unpopular well until the late 1950s. During the war, however, Turing leaned heavily on Bayesian reasoning in his statistical techniques for deciphering the naval Enigma codes. The irony is that while Fisher had once judged Turing's mathematics with skepticism, Turing's secret wartime breakthroughs relied on ideas Fisher would have firmly rejected, though Fisher never knew it at the time due to secrecy of Turing's work at Bletchley Park.

When Alan Turing applied for a research fellowship at King's College in 1934, his dissertation was sent to two external examiners. One of them was Ronald A. Fisher, already famous as one of the most influential statisticians of the century. Fisher did not take to the subject matter at all — he thought it was a dull and “positively repellent” area of analysis that had been worked over thoroughly by continental mathematicians. But as he read further, he could not ignore that Turing's treatment was original, elegant, and showed great technical mastery. In his final report, Fisher admitted that the work bore the mark of “a first-class candidate” and even praised Turing's “taste” and “virtuosity” in mathematical demonstration. That endorsement, even if reluctant, played a crucial role in Turing being awarded the fellowship and securing his place in the Cambridge academic world.

The irony comes later. Fisher was the leading opponent of Bayesian methods — dismissing them as heretical and pushing frequentist statistics as the only legitimate approach. His authority made Bayesian reasoning unfashionable in Britain for decades. Yet, during the Second World War, Turing developed Banburismus and other cryptanalytic methods at Bletchley Park that relied heavily on Bayesian reasoning — calculating probabilities from incomplete evidence to guide the breaking of German naval Enigma. In other words, Turing’s secret wartime triumphs were grounded in the very statistical philosophy Fisher most disliked. Because of the secrecy surrounding Bletchley’s work, Fisher never knew that the young man whose dissertation he had once judged so harshly had gone on to apply Bayesian methods to one of the most consequential problems of the century.

Prof. R.A. Fisher’s Further Report: I have no hesitation in judging Turing’s thesis the work of a first-class candidate. He seems to have thought out his own methods. Finally in reading through his paper, I formed a very high opinion of his taste, virtuosity would not be too strong a term, in the art of framing conclusive mathematical demonstrations.

When Turing applied for a fellowship at King’s College in 1934, his dissertation had to be judged by external experts. One of them was Ronald A. Fisher. Fisher didn’t like the topic much, calling it unattractive, but he still recognized that Turing’s approach showed real originality and ability. In spite of his misgivings, he recommended that Turing was a candidate of very high caliber.

The inquisition into Bayes’s heresy was led by none other than Professor R.A. Fisher, the Cambridge authority on statistics, he who had said that the subject Alan Turing had chosen for his fellowship dissertation was ‘positively repellent’.

Charles S. Peirce and I.J. Good

In *The Probability of Induction* (1878), Charles Sanders Peirce provided one of the earliest formulations of what later became known as the *weight of evidence*. He argued that belief should grow in proportion to the strength of independent arguments.

Charles Sanders Peirce wanted a way to measure how strongly independent arguments support a hypothesis — what he called the intensity of belief.

He reasoned like this:

“If two independent arguments both support the same conclusion, then the total belief should be the sum of the beliefs from each.”

That passage is one of the earliest known formal definitions of weight of evidence as an additive quantity in log-odds space, nearly seventy years before Turing made the same principle operational for Bayesian intelligence and statistical inference.

He then added:

“Since independent arguments multiply their chances, the feeling of belief should be proportional to the logarithm of the chance.”

The rule for the combination of independent concurrent arguments takes a very simple form when expressed in terms of the intensity of belief, measured in the proposed way:

“Take the sum of all the feelings of belief which would be produced separately by all the arguments pro, subtract from that the similar sum for arguments con, and the remainder is the feeling of belief which we ought to have on the whole. This is a proceeding which men often resort to, under the name of balancing reasons.”

Peirce observed that when independent pieces of evidence each favor a hypothesis to some degree, our total belief should combine additively in the scale of feeling or belief — even though, mathematically, probabilities themselves combine multiplicatively. He proposed that the appropriate “thermometer” for belief intensity must therefore be the logarithm of the odds (chance).

Almost a century later, I.J. Good re-examined Peirce’s argument in his short 1981 note “An Error by Peirce Concerning ‘Weight of Evidence’” [2].

Good agreed that Peirce’s intuition was brilliant but showed that the algebra behind it was wrong.

Peirce had written the independence of evidence as:

$$P(H \mid E, F) = P(H \mid E) P(H \mid F) \quad (1.29)$$

which implies

$$O(H \mid E, F) = O(H \mid E) O(H \mid F) \quad (1.30)$$

where $O(H)$ denotes the odds of the hypothesis H . That equation holds only if the prior odds $O(H) = 1$, i.e. a 50% prior probability, an implicit assumption Peirce never stated.

Translating to today’s methodology, Peirce effectively summed two weights of evidence without including any prior base rate, the intercept term, representing the prior odds before any evidence is observed.

Good pointed out that the correct independence condition is not on the posteriors but on the likelihoods:

$$P(E, F \mid H) = P(E \mid H) P(F \mid H) \quad (1.31)$$

$$P(E, F \mid \bar{H}) = P(E \mid \bar{H}) P(F \mid \bar{H}) \quad (1.32)$$

From Bayes’ theorem,

$$O(H \mid E, F) = O(H) \times \frac{P(E \mid H)}{P(E \mid \bar{H})} \times \frac{P(F \mid H)}{P(F \mid \bar{H})}. \quad (1.33)$$

Taking logarithms gives the additive law that Peirce intuited:

$$\log O(H \mid E, F) = \log O(H) + \log \frac{P(E \mid H)}{P(E \mid \bar{H})} + \log \frac{P(F \mid H)}{P(F \mid \bar{H})}. \quad (1.34)$$

Each term is a weight of evidence — the log-likelihood ratio contributed by one piece of data.

This mathematical framework, refined over more than a century from Peirce’s initial insight through Good’s correction, provides the theoretical foundation for modern credit scoring systems. The same principles that Peirce intuited for combining evidence now power the algorithms that process millions of loan applications, demonstrating the enduring value of fundamental mathematical insights.

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